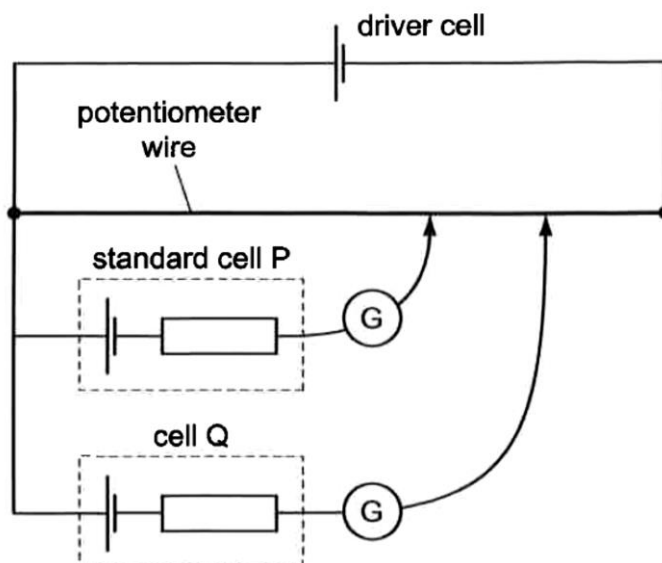


- 16** A potentiometer circuit is set up in which there is a standard cell P and a cell Q whose e.m.f. is to be determined. The two cells are used simultaneously, as shown below.



Is the value of the e.m.f. of Q, obtained by taking the two readings simultaneously, accurate and why?

- A** No, because cells P and Q will affect each other.
- B** No, because the internal resistances of P and Q will be different.
- C** Yes, because the e.m.f. of the driver cell is less than the e.m.f. of cell P or cell Q.
- D** Yes, because when readings are taken there is no current in either cell P or cell Q.

16	D	This is the working principle of potentiometer, i.e., both are at balanced lengths.
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17 A positive charge is placed at D and a negative charge is placed at Q. The electric potential